

C-1



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR SECOND SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE (BASIC SCIENCE)
INDUSTRIAL CHEMISTRY, AGRONOMY, HORTICULTURE,
ANIMAL SCIENCE, SOIL SCIENCE & PLANT NUTRITION,
AGRICULTURE EDUCATION AND EXTENSION,, EDUCATION
SCIENCE, PHARMACEUTICAL SCIENCE, MEDICAL LABORATORY
SCIENCE, MEDICAL BIOTECHNOLOGY, NUTRITION AND
DIETETICS & ECOTOURISM, HOTEL AND INSTITUTIONAL
MANAGEMENT WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

SCH 103: BASIC ORGANIC CHEMISTRY

Date: 12th July, 2018

Time: 12.00 - 3.00pm

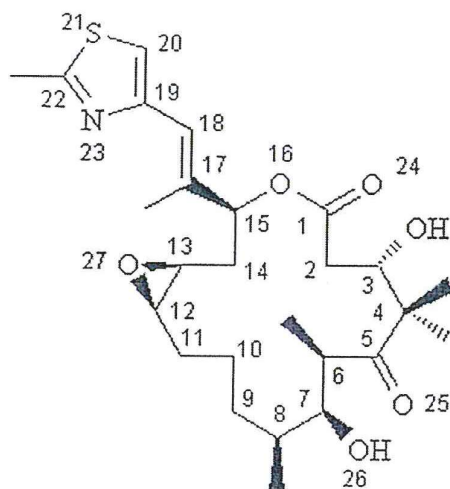
INSTRUCTIONS:

- Answer ANY FIVE Questions.



QUESTION ONE

- (a) Study the structure of epothilone A given below and answer the questions that follow:



- (a) Give the ground state electronic configuration of the following atoms in the molecule epothilone A.
- (i) O
 - (ii) N
 - (iii) S
- (b) What is the hybridization of the following atoms in epothilone A?

N23

O26

O24

C17

[2 marks]

(c) What type of bonding molecular orbitals make up the bonds between:

C17-C18

C21-C22

C3-C4

C13-O27

[2 marks]

(d) Arrange the following bonds in epothilone A in order of decreasing length:

C17-C18, C15-C17, C14-C15.

[2.5 marks]

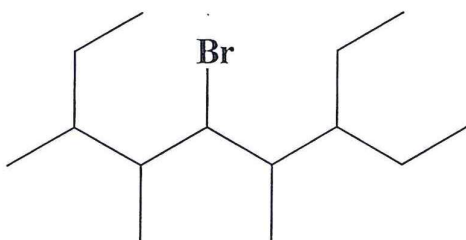
(e) Explain the order of decreasing length in (d) above.

QUESTION TWO

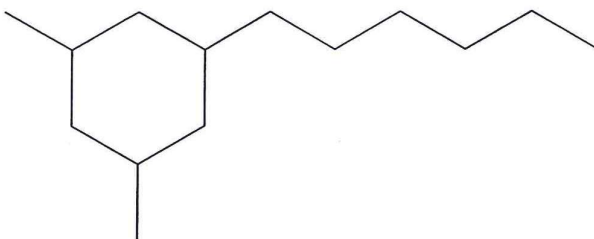
a) Name the following compounds using the IUPAC system.

[7marks]

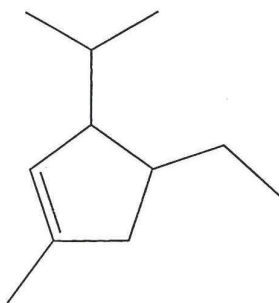
i)



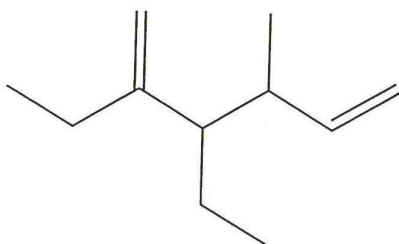
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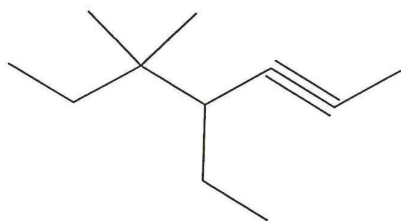
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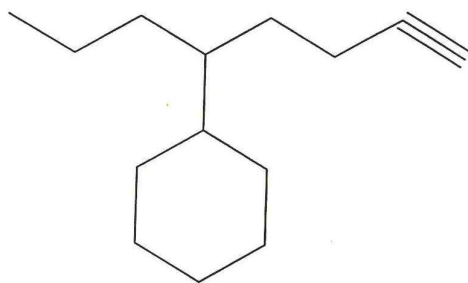
iv)



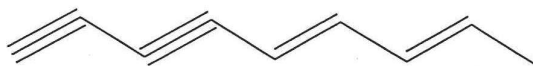
v)



vi)



vii)



b) Draw the structures of the compounds given below. [7marks]

- i) Chlorocyclopentane
- ii) 1-cyclohexyl-2,5-dimethylhexene
- iii) 5-methyl-3-propylhexene
- iv) 5-methyl-5-phenyldecyne
- v) 3-ethyl-4,6-dimethyloct-1,5-diene
- vi) 2,3-diethyl-1,3-heptadien-6-yne
- vii) 4-(1,3-dibromobutyl)-3-ethyltetradecane

QUESTION THREE

(a) (i) Define the term heat of reaction [1 mark]

(ii) Using the homolytic bond dissociation energies given below, calculate H for the reaction of ethane with bromine.

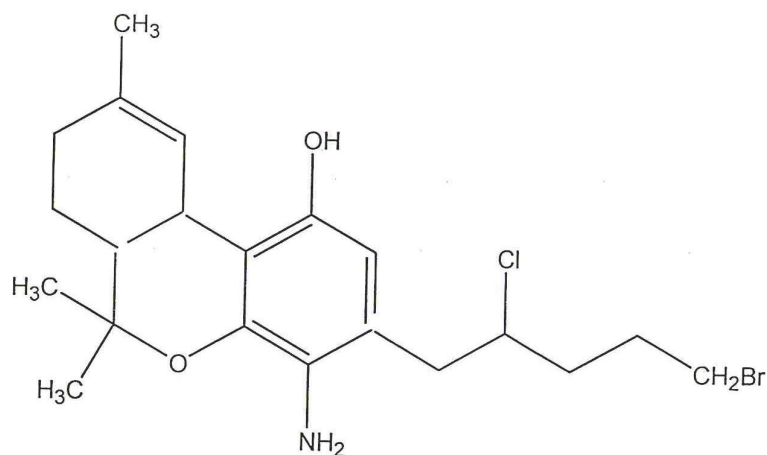
Comment on the nature of the reaction.



[3 marks]

(b) (i) What is degree of unsaturation? [1 mark]

(ii) Calculate the index of hydrogen deficiency for the molecule below:



[3 marks]

- (c) (i) Describe simple chemical tests that would distinguish between CH_3Cl and CH_3Br

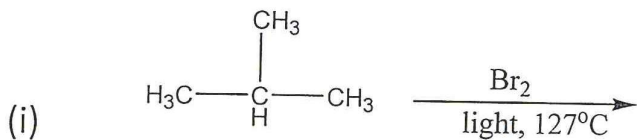
[2 marks]

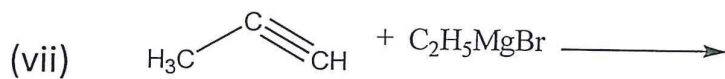
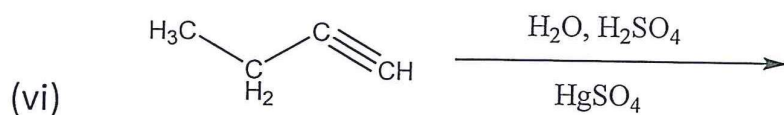
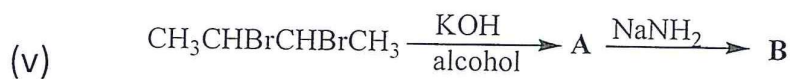
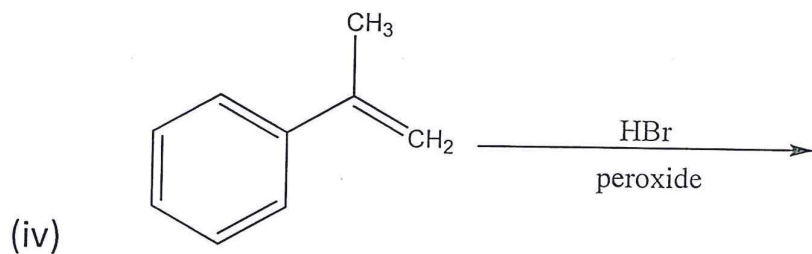
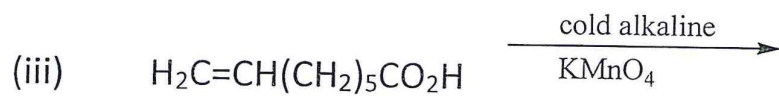
- (ii) The widely used antibiotic, penicillin G, gave the following elemental analysis:

C 57.45%, H 5.40%, N 8.45%, S 9.61%. The molecular weight of penicillin G is 334. Assuming that no other elements except oxygen are present, calculate the empirical and molecular formulae of penicillin G.

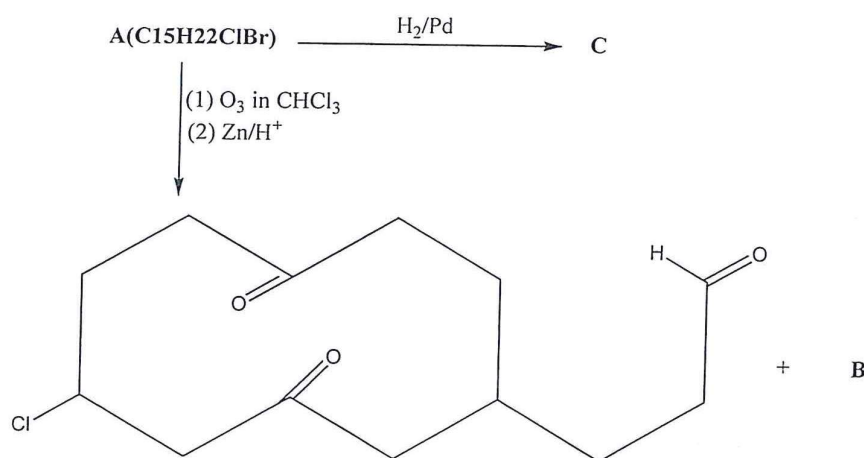
QUESTION FOUR

(a) Complete the following reactions:





(b) Study the reaction scheme shown below and answer the questions that follow:



Propose structures of A through C.

(c) When the alkene 3,3-dimethyl-1-butene is treated with hydrogen iodide, a mixture of products are formed, namely 3-iodo-2,2-dimethylbutane and 3-iodo-2,3-dimethylbutane.

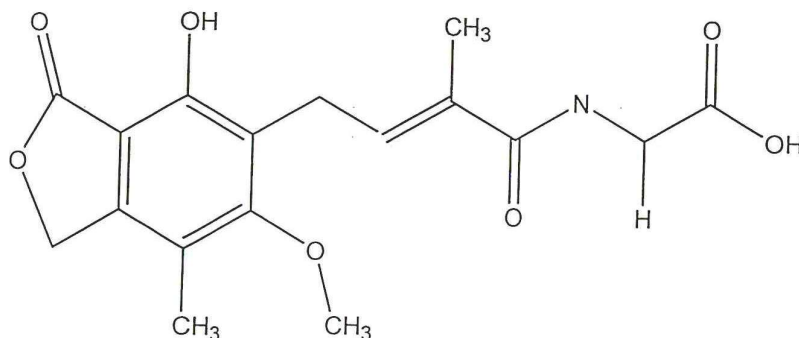
(i) What does the formation of 3-iodo-2,3-dimethylbutane suggest?

(ii) Propose the likely step leading to the formation of the two products.

QUESTION FIVE

(a) Below is the structure of the antiviral drug. Identify and name all the functional groups present in the compound

[7 marks]



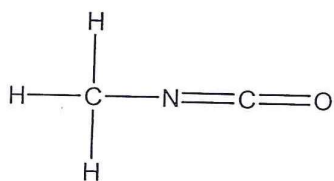
(b) Distinguish between functional group isomerism and position isomerism. [2 marks]

(c) Draw and name all the possible structural isomers of the hydrocarbon C_4H_8 . [3marks]

(d) Give the functional group of the isomers identified in 5c above. [2 marks]

QUESTION SIX

- (a) Draw a diagram showing all of the atomic orbitals in the molecules shown below. Indicate the orbitals that overlap to form sigma (δ) bonds, those that overlap to form pi (π) bonds and those which hold lone pair of electrons. Do not draw the back lobes of the hybrid orbitals.



[14 Marks]



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DEGREE OF BACHELOR OF SCIENCE IN ECOTOURISM, HOTEL
AND INSTITUTION MANAGEMENT WITH INFORMATION
TECHNOLOGY**

CITY CAMPUS

SCH 103: BASIC ORGANIC CHEMISTRY

Date: 13th July, 2018

Time: 9.00 - 12.00 noon

INSTRUCTIONS:

- Answer ANY FIVE questions



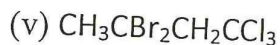
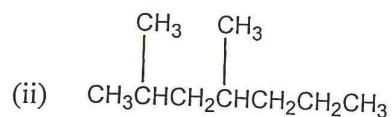
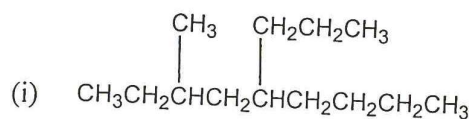
QUESTION ONE

Combustion of 6.51mg of a hydrocarbon compound gave 20.47 mg of carbon dioxide and 8.36 mg of water. The molecular weight was found to be 84. Calculate

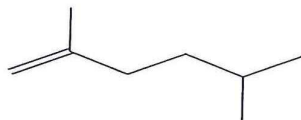
- (a) The percentage composition
 - (b) Empirical formula
 - (c) Molecular formula of the compound
 - (d) Nicotine has the molecular formula $C_{10}H_{14}N_2$. What is the index of hydrogen deficiency for nicotine?
- [14 marks]

QUESTION TWO

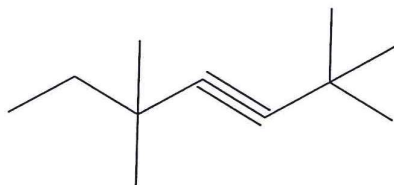
- (a) Give IUPAC names for the following structures.



(vi)



(vii)



[7 marks]

(b) Draw the structures corresponding to the following names.

(i) 3-Chloro-4,4-dimethyl-1-nonen-6-yne

(ii) 3-Ethyl-5-methyl-1,6,8-decatriyne

(iii) 3,4-Dimethylcycloheptyne

(iv) 2-Ethyl-3,4-dimethyl-1,4-pentadiene

(v) 3,5-Heptadien-1-yne

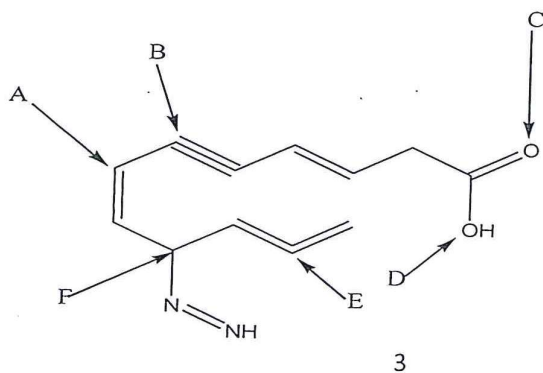
(vi) 7,7-dimethyl-6-ethyl-6-propyldecane

(vii) 1,2-dimethylcyclobutane

[7 marks]

QUESTION THREE

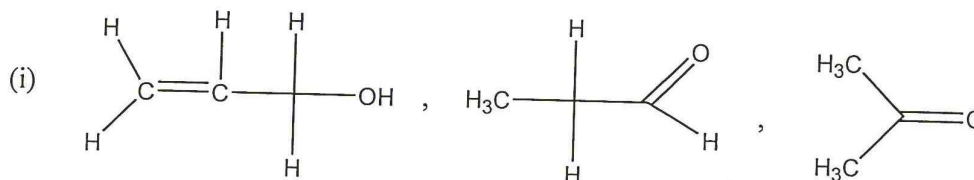
Use the structure shown below to answer the following questions:

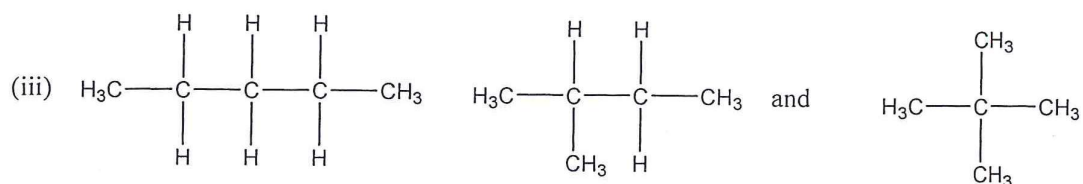
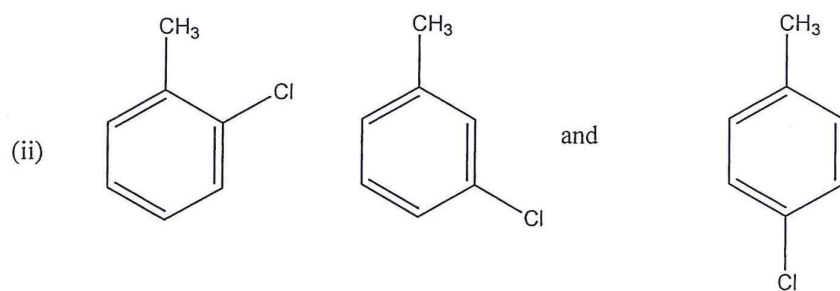


- (a) Identify the hybridizations labeled
- A
B
C
D
E
F
- (b) How many δ and π bonds are present in the nitrogen-nitrogen bond shown in the molecule
- (c) What are the orbitals that make up a π bond in a carbon-carbon double bond
- (d) Identify the total number of δ and π bonds in this molecule
- (e) What are the orbitals that make up a carbon-carbon σ bond
- (f) What orbitals make up the 3rd shell
- (g) Give the electronic configuration of phosphorus (15 electrons)
- (h) Identify the valence electrons in phosphorus [14 Marks]

QUESTION FOUR

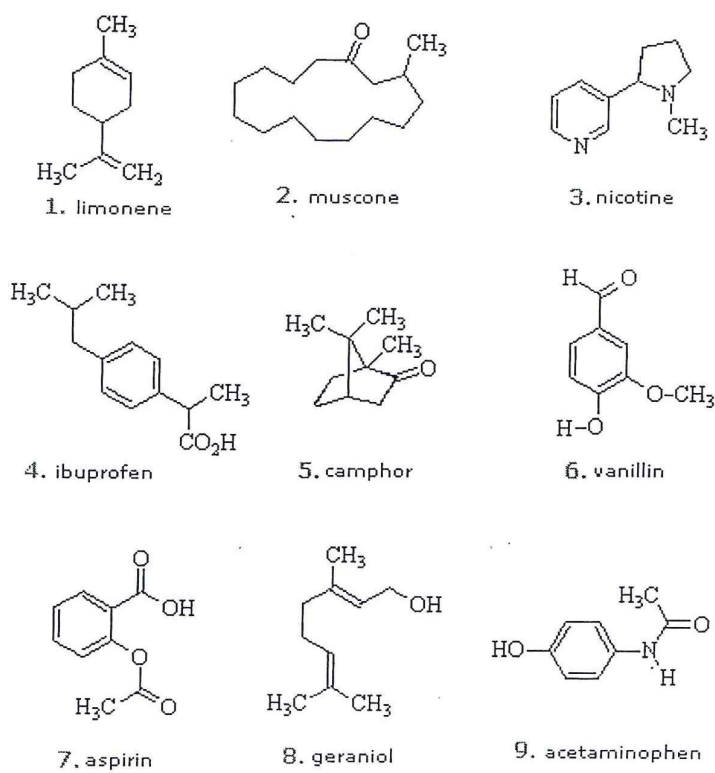
- (a) Identify the following sets of constitutional isomers as chain, position or functional group isomerism.





[3 Marks]

(b) The compounds whose structures are shown below incorporate a variety of functional groups.



Identify the compounds that have the following specific functional groups. For each compound that has the designated functional group, use appropriate number to represent the structure.

- (i) Which structures have a carboxylic acid group?
- (ii) Which structures have a ketone carbonyl group?
- (iii) Which structures have carbon-carbon double bonds?
- (iv) Which structures have ether group?
- (v) Which structures have aromatic ring?

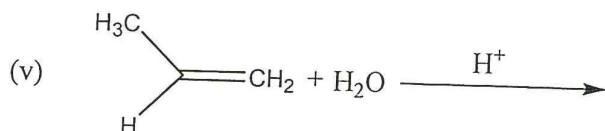
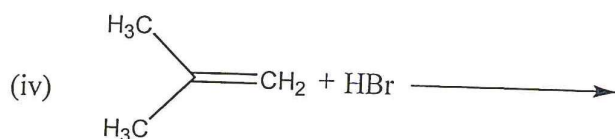
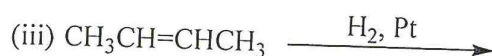
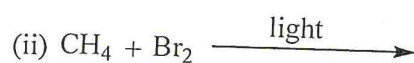
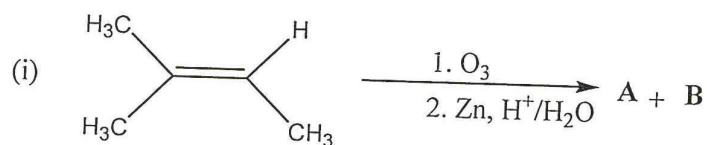
[6 marks]

(c) Carotene is a valuable dietary source of vitamin A.

- (i) Mention some of the sources of carotene
- (ii) What are the main uses of vitamin A? [5 Marks]

QUESTION FIVE

(a) Complete the following reactions:





[7 Marks]

(b) Define the following terms

- (i) Functional group
- (ii) Hybrid orbitals
- (iii) Isomers
- (iv) Markovnikov's rule
- (v) Index of hydrogen deficiency

[5 Marks]

(c) Name four kinds of organic reactions

[2 Marks]

QUESTION SIX

- (a) Give the general formula of an alkane
- (b) What are the different types of structural isomers, and where necessary give an example of each type.

(c) What are the different types of organic chemical reactions?

(d) Draw a complete line bond and electron-dot formulae for ethanoic acid.

(e) What is the hybridization of oxygen atom in CH_3OH ? Draw the Lewis structure of this compound.

[14 Marks]



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UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF
EDUCATION WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

SCH 104: BASIC ANALYTICAL CHEMISTRY

Date: 9th July, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS

- Answer ANY FIVE questions



MASENO UNIVERSITY EXAMINATIONS

SCH 104: BASIC ANALYTICAL CHEMISTRY

Instruction

Answer any five questions

Useful information

Atomic masses for S, O, Al, H, Fe, N, Cu, I, Ag and Cr are 32, 16, 27, 1, 56, 14, 63.5, 127, 108 and 52 respectively.

1. A mass of 2.246 g of aluminium sulphate is used to do an experiment. By clearly showing your working, determine the number of moles of:

- a) aluminium sulphate. (2.5 marks)
- b) aluminium ions and hence their mass. (3.5 marks)
- c) sulphate ions (2 marks)
- d) sulphur atoms that make up this sulphate ions and hence their mass. (3 marks)
- e) oxygen gas molecules that make up this mass and hence their mass. (3 marks)

2. a) Acid-base reaction is one type of a chemical reaction commonly referred to as neutralization reaction. Given below are two examples of acid-base reactions.

(I) Strontium hydroxide (aq) + Perchloric acid (aq) $\rightarrow$

(II) Barium hydroxide (aq) + Sulfuric acid (aq) $\rightarrow$

For each chemical reaction above write the

- (i) complete balanced molecular equation
- (ii) total ionic equation
- (iii) net ionic equations and
- (iv) identify the spectator ions

(12 marks)

b) What information can be obtained from the net ionic equation that cannot be obtained from the molecular equation?

(2 marks)

3. a) Distinguish between oxidizing and reducing agent. (2 marks)
- b) Distinguish between a Lewis acid and a Bronsted acid. (2 marks)
- c) The pH of human arterial blood is 7.40. What is the hydronium-ion concentration? (2 marks)
- d) A litre of a solution saturated at 25°C with calcium oxalate, CaC_2O_4 , is evaporated to dryness, giving a 0.0061 g residue of CaC_2O_4 . Calculate the solubility product constant for this salt at 25°C. (4 marks)
- e) The concentration of calcium ion in blood plasma is 0.0025 M. If the concentration of oxalate ion is 1.0×10^{-7} M, determine if the calcium oxalate will precipitate? K_{sp} for calcium oxalate is 2.3×10^{-9} . (4 marks)
4. (a) What is meant by the following terms as applied in analytical chemistry?
- (i) Ultimate analysis
 - (ii) Complete analysis
 - (iii) Analyte
 - (iv) Precision
- (4 marks)
- (b) Aluminium solid granules measuring 1.36 g is reacted with 1.5 g of oxygen gas to form aluminium oxide. Assuming the reaction reached completion:
- (i) Give a stoichiometric equation of the reaction. (1 mark)
 - (ii) Determine the number of moles of aluminium oxide formed in this reaction. (2.5 marks)
 - (iii) Deduce the limiting reagent in this reaction. (1.5 marks)
- (c) A number of chemical instruments utilising various techniques are used to do chemical analysis of chemical samples.

(i) List any three of these instruments and give the technique employed in each instrument. (3 marks)

(ii) Mention any two major applications of analytical chemistry. (2 marks)

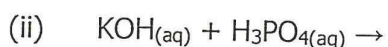
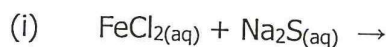
5. a) Define the following

(i) Salt

(ii) Net ionic equation (2 marks)

b) Distinguish a weak electrolyte from a strong electrolyte. (2 marks)

c) Predict whether mixing each pair of the following solutions will result in the formation of a precipitate, and so, identify the precipitate.



d) A mixture contains only NaCl and $\text{Fe}(\text{NO}_3)_3$. A 0.356g sample of this mixture is dissolved in water, and then an excess of NaOH is added, resulting in formation $\text{Fe}(\text{OH})_3$ precipitate. If 0.104 g of $\text{Fe}(\text{OH})_3$ is collected.

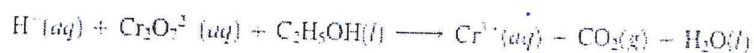
(i) Write the ionic equation for the formation of iron (III) hydroxide. (1 mark)

(ii) Determine the mass of iron in the original sample (2 marks)

(iii) Calculate the mass of $\text{Fe}(\text{NO}_3)_3$ in the original sample (2 marks)

(iv) Deduce the % mass of $\text{Fe}(\text{NO}_3)_3$ in the original sample. (2 marks)

6. a) Potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) is a bright orange compound that can be reduced to a blue-violet solution of Cr^{3+} ions. Under certain conditions, $\text{K}_2\text{Cr}_2\text{O}_7$ reacts with ethyl alcohol as follows:



Balance this equation using the half-reaction method (4 marks)

b) Consider the following reactions (I) and (II):

(I) Aqueous potassium chloride is added to aqueous silver nitrate to form a silver chloride plus potassium nitrate.

(II) Aqueous potassium hydroxide is mixed with aqueous iron (III) nitrate to form iron(III) hydroxide and potassium nitrate.

For each of these reactions, write the formula equation, the complete ionic equation, and the net ionic equation. (4 marks)

c) The K_{sp} value for copper (II) iodate, $\text{Cu}(\text{IO}_3)_2$, is 1.4×10^{-7} at 25°C . Calculate its solubility at 25°C . (3 marks)

d) Calculate the number of moles of Ag_2CrO_4 that will dissolve in 1.00 L of 0.010 M K_2CrO_4 solution. K_{sp} for $\text{Ag}_2\text{CrO}_4 = 9.0 \times 10^{-12}$. (3 marks)



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FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF MASTER OF SCIENCE IN CHEMISTRY

MAIN CAMPUS

SCH 809: INTRODUCTION TO STATISTICAL METHODS IN CHEMISTRY

Date: 18th July, 2018

Time: 9.00 - 12.00 pm

INSTRUCTIONS:

- Attempt ANY FIVE Questions.



SCH 809: INTRODUCTION TO STATISTICAL METHODS IN CHEMISTRY**Answer any five questions****Time: Three hours****1. a) Explain the following terminologies:**

- i. Mean (1 marks)
- ii. Range (2 marks)
- iii. Quartile (2 marks)
- iv. Median (2 marks)
- v. Decile (2 marks)

b) In a study to evaluate the differences in clonal mature leaf nitrogen with region of production (Kwach, Owuor, Kamau, & Wanyoko, J.K. *Asian Journal of Biological and Life Sciences*, **1(1)**, 8-18 (2012)), the following were data (% nitrogen) were obtained..

Clone	Timbilil	Kipkebe	Kangaita
TRFK7/9	3.62	3.45	3.25
TRFK303/259	3.62	3.67	3.55
TRFK303/1199	3.18	3.28	2.67
TRFK54/40	3.42	3.51	2.97
TRFK31/8	3.11	3.62	2.76
BB35	3.10	3.31	2.91
TRFK6/8	3.55	3.64	3.15
TRFK31/27	3.54	3.40	3.30
TRFK12/12	3.44	3.46	3.39
TRFK303/999	3.26	3.32	3.19
AHPS15/10	3.30	3.66	2.73
TRFK57/15	3.74	3.65	3.28
TRFK56/89	3.25	3.32	3.30
TRFK12/19	3.39	3.34	3.21
TRFK11/26	3.57	3.39	3.40
STC5/3	3.68	3.58	2.89
TRFK7/3	3.16	3.25	3.25
TRFK303/577	3.41	3.49	3.04
EPKTN14-3	3.64	3.10	3.17
TRFK2xI/4	3.43	2.80	3.26

- i. Draw box plots on one graph of the presented data (5 marks)
- ii. Compare the distribution. (2 marks)
- iii. Explain the difference between a box plot and a box graph (2 marks)
- iv. Using any one location draw a box graph of the data (2 marks)

2. Withering is an important step in black tea processing, during which leaf loses moisture content and becomes flaccid and easy to macerate. This process is also referred to as physical wither. However, the process is also accompanied by biochemical changes in the leaf, which influence the activities of the enzymes responsible quality. Such biochemical changes are known as chemical wither. Owuor and Obanda, *International Journal of Tea Science*, **13** (1&2), 72-79 (2017), studied the influence of degree of hard

wither followed by rehydration and withering duration on the theaflavins ($\mu\text{mole/gm}$) content of clone TRFK 6/8 tea. The replicated mean results were as follows:

	Wither duration (hours)	Theaflavins ($\mu\text{mole/gm}$)
Chemical wither	14	20.94
	22	25.01
	30	23.15
	38	22.23
	44	21.24
Slow hard physical	14	19.17
	22	21.49
	30	19.78
	38	18.77
	44	16.97
Rehydrate hard physical	14	22.84
	22	22.81
	30	22.49
	38	23.10
	44	17.49

- Identify the study variables and classify them as independent and dependent variables **(3 marks)**
- What could be the possible experimental design of the trial? **(2 marks)**
- The CV (%) of the analysis was calculated as 11.76 while LSD ($p \leq 0.05$) values were 2.91 and 3.26 for wither type and duration, respectively. Construct a table that can help you interpret the results and calculate treatment means. **(9 marks)**
- Briefly discuss the effects of withering duration on theaflavins levels. **(4 marks)**
- Give one recommendation on methods of withering. **(2 marks)**

3. a) An agricultural chemistry student surveyed the infestation of sugarcane farms by nematodes in the Nyando Sugar Belt. The farms were either managed by the owners or trained managers. Several farms were visited and were classified as nematode infested or nematode free. The distribution of the farms by presence or absence of nematodes were as follows:

Management	Nematode infested	Nematode free
Owner	1600	900
Managers	800	500

- Construct a sample space and probability function for this experiment. **(2 Marks)**
- Is the chance that sugarcane farm was nematode infested dependent on manager? **(8 Marks)**

b) An environmental chemistry student evaluated the extent lead contamination of in fish eaten in the busy Lwangni Hotel in Kisumu. Out of the customers that ate fish in the hotel over a period of one week, 45% were females. Exactly 40% of the fish was lead contaminated.

- Construct a sample space and probability function for this evaluation. **(2 Marks)**

- ii. Is the chance that a customer ate lead contaminated fish dependent on gender?
(8 Marks)

4. The accompanying table shows variations in the levels (as ratio of gas chromatographic peak are of the compound to that of an internal standard) of four key volatile flavour compounds in black tea due to fermentation durations (Owuor, Obanda, Nyirenda and Mandala, *Food Chemistry* 108, 263-271 (2008)).

Volatile flavour compound	Fermentation duration (Mins)				
	30	50	70	90	110
E-2-Hexenal	1.87	2.03	2.56	2.41	2.77
Hexanal	0.28	0.36	0.45	0.45	0.57
Linalool	0.85	0.69	0.61	0.54	0.46
Geraniol	1.74	1.67	1.51	1.47	1.47

- Construct scatter plot showing how levels of each E-2-hexenal and linalool vary with fermentation duration.
(3 Marks)
- Discuss your findings.
(1 Mark)
- Construct a scatter matrix using these four volatile flavour compounds
(14 Marks)
- What observations can you make about the relationships between pairs of variables?
(2 Marks)

5. The variations in the levels of leaf potassium levels (%) in mature leaf and two leaves and a bud of twenty popular tea clones grown in Kenya are as follows:

Clone	Mature leaf	2 leaves and a bud
TRFK7/9	1.33	2.03
TRFK303/259	1.69	2.13
TRFK303/1199	1.52	2.16
TRFK54/40	1.44	2.05
TRFK31/8	1.62	2.06
BBK35	1.19	1.99
TRFK6/8	1.52	2.12
TRFK31/27	1.71	2.17
TRFK12/12	1.69	2.05
TRFK303/999	1.70	2.22
AHPS15/10	1.80	2.21
TRFK57/15	1.45	2.27
TRFK56/89	1.31	2.24
TRFK12/19	1.45	2.12
TRFK11/26	1.34	2.12
STC5/3	1.19	1.99
TRFK7/3	1.46	2.19
TRFK303/577	1.55	2.19
EPKTN14-3	1.64	1.97
TRFK2xI/4	1.72	2.28

- What is a simple linear regression? **(2 marks)**
- Develop a linear regression equation for the levels of potassium in mature leaf and in two leaves and a bud of clonal tea **(8 marks)**
- Calculate the linear correlation coefficient for the levels of potassium in mature leaf and in two leaves and a bud of clonal tea. **(6 marks)**
- How much of the relationship was explained by your equation? **(4 marks)**

6. a) Define the following terminologies:

- i) A data set (1 Mark)
- ii) A statistic (1 Mark)
- iii) Data analysis (1 Mark)
- iv) Effective graph (1 Mark)
- v) Population (1 Mark)
- vi) Sample (1 Mark)
- vii) Statistical inference (1 Mark)
- viii) Trimmed mean (1 Mark)
- ix) Residual (1 Mark)
- x) Symmetric distribution (1 Mark)

b) The nitrites levels (ppb) in water from springs flowing into Rever Amala in the Mara Basin (Owuor, J.J. MSC Thesis, Maseno University 2013) are as follows

Spring	Kebenet	Chepudonge	Motiok	chepkesoi	Kapangas	Kapangas	Ndong Ndong	Siongiroi
Nitrites (ppb)	3.070	4.560	11.950	10.490	5.910	4.160	5.950 ^c	2.530 ^h

- i. How many variables were in the survey? (2 marks)
- ii. Classify the variables (2 marks)
- iii. Determine the standard deviation of the nitrite in the spring waters. (6 marks)



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF MASTER OF SCIENCE IN CHEMISTRY

MAIN CAMPUS

SCH 811: ANALYSIS OF INDUSTRIAL EFFLUENTS

Date: 17th July, 2018

Time: 9.00 - 12.00 pm

INSTRUCTIONS:

- Attempt ANY THREE Questions.



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SCH 811: Analysis of Industrial Effluents (60 Marks)

Instructions: Answer any three questions

1.

- a) Dissolved oxygen is one of the most important parameters analysed for in industrial effluents.
 - i. Discuss the factors that affect the level of dissolved oxygen in an unpolluted stream [6 marks]
 - ii. How could pollutants lead to the depletion of dissolved oxygen in a stream? [2 marks]
 - iii. How does super saturation arise in a water body? [1 mark]
- b) The Winkler test and electrode methods have been used for a long time for the analysis of dissolved oxygen in industrial effluents and environmental waters
 - i. Highlight and explain the major steps in the Winkler test [5 marks]
 - ii. What are some of the weaknesses of the Winkler test? [3 marks]
 - iii. Explain how some of the weaknesses of the method can be overcome [3 marks]

2.

- a) Detergents are commonly detected in industrial effluents.
 - i. Outline the method of analysis for anionic detergents [2 marks]
 - ii. What are the possible consequences of the presence of detergents and other surfactants in natural waters? [5 marks]
 - iii. Outline how you would determine the total phosphorus content of a sample spectrophotometrically? [3 marks]
- b) Presence of high amounts of ammoniacal-N and other forms of nitrogen in streams and rivers is usually due to waste water effluents.
 - i. What are some of the negative effects of ammoniacal-N? [2 marks]
 - ii. Using equations show how nitrification could occur in waters contaminated with ammoniacal-N [2 marks]
 - iii. Explain how you can determine the three nitrogen forms i.e. ammoniacal-N, total oxidized nitrogen and organic-N spectrophotometrically [6 marks]

3.

- a) Alkalinity is often determined in environmental water samples.
 - i. What is alkalinity? [1 mark]
 - ii. With the aid of chemical equations, show how the alkalinity of a water system buffers it against changes in pH. [4 marks]
- b) Briefly outline how you would determine the following parameters in an industrial effluent: [4marks]
 - i. Total Organic Carbon (TOC)
 - ii. Turbidity

- c) What is the importance of analyzing for TOC and turbidity in the water of a receiving stream? [2 marks]
- d) When analyzing for industrial pollutants in the water of a receiving stream, why is it common to also analyze for the pollutants in the sediments of the stream? [3 marks]
- e) Biological oxygen demand (BOD) is a routine parameter that is determined in water quality analysis.
 - i. What is BOD and how is it determined? [3 marks]
 - ii. The BOD test has a number of disadvantages and limitations. In spite of this, it is still widely used. What is the significance and relevance of the BOD test? [3 marks]

4.

- a) In the Kjeldahl determination of organic nitrogen, concentrated sulphuric acid, copper (II) chloride, potassium sulphate and Nessler's solution (KI.HgI_2) are used. What is the role of each of these compounds in the analysis? [4 marks]
- b) Hardness of water is often determined in effluents and receiving waters.
 - i. What causes water hardness and what is its significance in natural waters? [2 marks]
 - ii. How would you establish temporary, permanent and total hardness in an industrial effluent by simple titration? [2 marks]
- c) An industrial effluent has been collected from a petroleum refinery. You are required to confirm whether or not it contains polycyclic aromatic hydrocarbons (PAHs). Outline and explain the steps you would take to determine the levels of PAHs in the effluent. (Include and explain all the steps from extraction to analysis) [7 marks]
- d) Mining activities result in release of heavy metals into receiving waters. Outline and explain how you collect and analyze for metal contaminants in a sediment sample, given that most of the metals are trace levels [5 marks]



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR SECOND SEMESTER EXAMINATIONS
FOR THE DEGREE OF MASTER OF SCIENCE IN
CHEMISTRY**

MAIN CAMPUS

**SCH 814: BIOCHEMICAL TECHNIQUES AND SEPARATION
METHODS**

Date: 20th July, 2018

Time: 9.00 - 12.00 pm

INSTRUCTIONS:

- Attempt ALL Questions.
- Requirements are Calculator and graph paper



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SCH 814: BIOCHEMICAL TECHNIQUES AND SEPARATION METHODS

Time: 3 hours

INSTRUCTIONS:

Answer question **ALL** questions

REQUIREMENTS: Calculator, graph paper.

QUESTION ONE [15 MARKS]

- a. Outline any three advantages of radioactivity as compared to other analytical techniques. [3 marks]
- a. Although the absorption spectrum is primarily determined by the chemical structure of the chromophore, environmental factors can affect $\lambda_{\max}$ and ϵ . Briefly describe the three environmental factors that affect $\lambda_{\max}$ and ϵ in an absorption spectrum. [6 marks]
- b. According to the sedimentation theory, the velocity (v) of a particle in a centrifugal field is given by the equations: $V = \omega^2 r m u (\rho_p - \rho_s) / f$; derive the expression above given that F_c = the centrifugal force, F_b = buoyant force, frictional force (F_f), m = mass of the particle, ω = angular velocity, r = distance from the axis, v = velocity of the particle, m_o = the mass of the displaced solution, f = frictional coefficient, u = partial specific volume of the particle ρ_p = density of particle and ρ_s = density of the solvent. [6 marks]

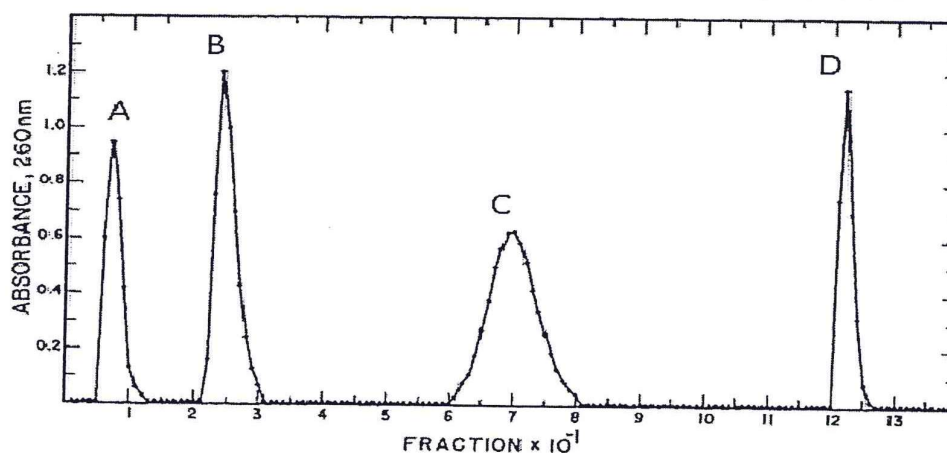
QUESTION TWO [15 MARKS]

- a. Differentiate the following terms as used in the separation of proteins: [6 marks]
 - i. Isoelectric point and chromatofocusing
 - ii. Chaotropic agents and detergents
 - iii. Ion exchange chromatography and gel filtration chromatography

- b. Draw a flow diagram showing the basic components of a high performance liquid chromatograph. [5 marks]
- c. Dehydrogenase activities are easily measured in spectrophotometric assays because of strong absorbance of reduced flavin nucleotides (NADH and NADPH) at 340 nm as compared to the oxidized nucleotides (NAD⁺, NADP⁺). The ϵ_{340} of either NADH or NADPH is 6.22×10^3 liter/cm·mole. Calculate the amount of glutamate dehydrogenase (GDH) activity expressed in moles glutamate formed per min per ml of protein using the following reaction parameters: the protein concentration in the sample is 0.23 mg/ml; 0.01 ml of sample was measured the final cuvette volume was 1 ml; the A_{340} decreased from 0.60 to 0.52 in 10 minutes; and the A_{340} of a blank (containing no enzyme) decrease from 0.60 to 0.59 in 10 minutes. GDH catalyzes the following reaction: α -ketoglutarate + NH₄⁺ + NADPH $\leftrightarrow$ glutamate + NADP⁺ [4 marks]

QUESTION THREE [15 MARKS]

- a. The following chromatogram was obtained during separation of a mixture of adenine nucleotides containing adenosine, adenosine monophosphate (AMP), adenosine diphosphate (ADP) and adenosine triphosphate (ATP) Using formate as counter ions in ion exchange chromatography. Study the chromatogram and answer the questions that follow.

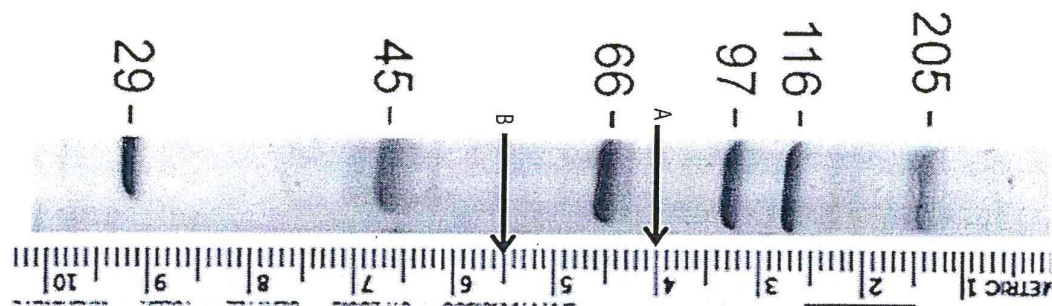


- i. Name the peaks A, B, C and D. (2 marks)

- ii. Explain the order of elution. (4 marks)
- b. Background radiations usually influence the measurements obtained during radioactivity in experiments. List any four sources of background radiations. [4 marks]
- c. Briefly describe how radioactivity is measured using liquid scintillation spectrophotometry. [5 marks]

QUESTION FOUR [15 MARKS]

- b. Differentiate DNA and RNA bases on their structures. [4 marks]
- c. In gel electrophoresis, molecular masses of proteins can be estimated by comparing the migration (relative mobility or R_f) of proteins of interest to standards of known size. You are provided with the following diagram showing the migration of proteins with different molecular weight on a gel. Two proteins samples (A and B) were separated under similar conditions and their migration distances are indicated on the gel. Showing the steps involved, determine the molecular weight of A and B. NB: MW is given in x1000 daltons. [11 marks]



mass (daltons)	log (mass)	distance migrated (mm)	R_f ($\div 101$ mm)
205,000	5.31	14.5	0.14
116,000	5.06	27	0.27
97,000	4.99	33	0.33
66,000	4.82	45	0.45
45,000	4.65	67	0.66
29,000	4.46	92	0.91

QUESTION FIVE [15 MARKS]

- a. Explain the role of the following steps during isolation of nucleic acids.
- i. Cell lysis: [3 marks]
 - ii. Enzymatic treatment: [2 marks]
 - iii. Phenol Extraction: [2 marks]
- b. A student was required to isolate iridoid glycosides from dry plant leaves of *Mangiferin indica* (mango leaves) using column chromatography. The student was provided with dichloromethane, petroleum ether, methanol and water as extraction solvents and dry mango leaves. Giving explanations outline the main steps the student followed to obtain the extract ready for iridoid isolation by column chromatography. [8 marks]



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR SECOND SEMESTER EXAMINATIONS
FOR THE DEGREE OF MASTER OF SCIENCE IN
CHEMISTRY**

MAIN CAMPUS

SCH 815: TREATMENTS OF INDUSTRIAL EFFLUENTS

Date: 16th July, 2018

Time: 9.00 - 12.00 PM

INSTRUCTIONS:

- Attempt any 4 questions , each question is 15 marks



SCH 815: TREATMENTS OF INDUSTRIAL EFFLUENTS**TIME: 3 HOURS****Instructions: Attempt any 4 questions, each question is 15 marks.**

Q1a) Define the following terms as they apply to waste treatments (6 marks)

- i) Environmental risk
- ii) Waste Energy factor
- iii) Selectivity of a process

Q1b) Incinerators are no longer considered as waste disposal sites but as energy recoveries. State the advantages and disadvantages of the incinerators. (5 marks)

Q1c) Waste management is considered a very dirty job. What should be the best protocols associated with the process of a waste treatment. (4 marks)

Q2a) Chemical precipitation is one of the methods commonly used to purify liquid wastes.

- i) State which any six (6) factors that affect rates of chemical precipitation (3 marks)
- ii) With a help of a diagram (log of concentration Vs pH) explain a theoretical solubility curve of a metal ions (5 marks)

Q2b) What is the solubility of PbCl_2 in 0.10 M NaCl? K_{sp} for PbCl_2 is 1.7×10^{-5} . (7 marks)

Q3) You are consulted to recommend to Kisumu City on solid waste management. Briefly outline your most considered inputs under the following topics;

- a) Categorization of the solid wastes (4 marks)
- b) Minimization of the waste load (4 marks)
- c) Precautions to observe during collection and transportation of the waste (5 marks)
- d) Disposal/management of the waste (2 marks)

Q4a) Explain how best you would handle a known radioactive element spillage in a drainage canal. **(9 marks)**

Q4b) Identify and describe the major classes and characteristics of wastes **(6 marks)**

Q5a) What is the main difference between nutrient removal and recovery? **(2 marks)**

Q5b) The water hyacinth in Lake Victoria is attributed to a lot of nutrient loads from industrial set-ups around the lake.

i) What are the likely causative elements responsible for the menace? **(4 marks)**

ii) As a scientist, how best would you address causes of this menace? **(6 marks)**

iii) What are the likely challenges you are likely to face when solving the above problem? **(3 marks)**

Q6 For centuries, humanity has been using activated carbon as the number one solution for environmental contaminations. However, it is now realized it cannot solve all the problems. Justify this observation. **(15 marks)**

.....**END**.....